Dungeons and Dragons Challenge Tier Forecast

Team Members and Responsibilities

Team Member/Lead: **Leo Foy**

Individual project, full responsibility. Communication between past and future me happens over Notion via notes and a project management database

Project Problem Description

It is often hard to know how challenging a monster is in Dungeons and Dragons. The company behind Dungeons and Dragons has the ability to do rigorous testing of how statistics influence a monster's general challenge. Dungeon Masters alone do not have this ability when making their own monsters. The Dungeon Masters guide, created by Dungeons and Dragons, even states that when modifying a creature or making your own, one shouldn't change certain stats because it can alter the challenge rating, providing no further guidance on the matter. Thus, a model that can forecast a monster's appropriate tier of play based upon the established trends found in existing monsters is useful to a Dungeon Master.

The I/O and AI Task

Input → User defined Monster Statistics

AI Task → Numerical Classification

Output → A Challenge Rating Tier forecast (based upon the official DnD tiers of play)

Data Set and Target

The data will be derived from a project that utilizes a well-made json file of the monsters I need, properly licensed. The size of this data set is about 322 monsters, a small section of which will be used for evaluation. There are a few important data points and variables that I

will need to parse through and use. The core monster statistics such as Strength will be a powerful indicator of challenge, as well as health and armor class. Predefined challenge ratings will be used for training. The target output of these statistics will be a tier of challenge that encompasses a small range of specific challenge ratings. The challenge rating data is imperfect in nature, and thus it may be more useful for a dungeon master to know whether the monster is generally appropriate for a certain tier of play rather than a specific numbered CR. Pre-processing of the data will be necessary for removing irrelevant monster information and separating records into training and evaluation sets.

The Initial Approach

My initial approach will utilize a Decision Tree (is a stat above a numerical threshold?)

Project Resources

- This project contains a json dataset of monsters with all of the information I will need: https://github.com/adkinn/srd-5.2.1?tab=License-1-ov-file
- A tutorial to help me get started with decision trees in python: https://youtu.be/YkYpGhsCx4c?si=aZJ8kIzm_C_jbHHS
- This resource explains how to evaluate a model, which I will need to know to test how reliable my model is at predicting: https://scikit-learn.org/stable/modules/model_evaluation.html
- A different solution to a similar problem, done with regression. This project could help me realize holes in my domain knowledge of Dungeons and Dragons: https://github.com/trevoriwen/build-a-monster

Proposed Project Timeline

Oct. 8 - Oct. 11	Define problem and prepare data and GitHub Setup
Oct. 26–Nov. 15	Explore/preprocess data and develop baseline / Develop and improve AI approach
Nov. 16 – Dec. 4	Evaluate, analyze, and refine system
Dec. 5 –Dec. 11	Finalize report, demo, and presentation

MVP

In order to reach a minimum viable product, the project must have a front-facing user interface that can interact with user input. Using the input (monster statistics), the internal AI must decide based on those what challenge tier the monster is (tiers 1-4). The tier of the monster must be output in a readable format.